

Urban Pulse - Smart Tourism & Cultural Intelligence Platform

STONE 3 - DEMAND INTELLIGENCE & VISITOR MOBILITY
ANALYTICS

TEAM - 1

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Problem Statement

Milestone 2 gave us visibility into where and when tourism demand concentrates, but stopped at aggregate footfall numbers. It couldn't answer operational questions: booking completion vs cancellation rates, entry queue congestion, how tourists move within a heritage precinct, or how sharply demand shifts with weather.

Milestone 3 closes this gap: we move from monument-level footfall to visit-level and trip-level intelligence, giving tourism boards the same operational granularity a ride-hailing platform has, applied to heritage tourism.



Problem Objectives

1 Hourly Demand Tracking
Track hourly visitor demand and identify peak arrival windows at heritage sites

2 Booking Intelligence
Measure booking completion, cancellation, and no-show rates

3 Queue Monitoring
Monitor entry queue and wait times to detect bottlenecks at marquee vs offbeat sites

4 Last-Mile Analytics
Analyze last-mile transport (buses, metros) within heritage precincts

5 Transit Mode Shifts
Understand how tourists switch transit modes and visit formats

6 Weather Sensitivity
Quantify how weather conditions shift demand between outdoor and indoor sites

Project Scope

In Scope

- Visit and Booking Intelligence dashboards (hourly demand, booking status, queue times)
- Last-Mile Heritage Circuit and Micro-Mobility analytics (buses/metro station utilization)
- Modal Substitution and Transit Shift analysis
- Weather Sensitivity and Demand Elasticity modeling
- Synthetic data generation for transaction-level booking and transport data

Out of Scope

- Executive-level accessibility index (Milestone 4)
- Final unified cross-module dashboard (Milestone 4)
- Real-time live data feeds (this milestone uses historical and simulated data)

Proposed Solution

Milestone 3 extends our Galaxy schema with three new Fact tables, split across four connected modules:



Module 1 - Visit & Booking Intelligence

Hourly demand, booking completion/cancellation, queue times



Module 2 - Last-Mile Heritage Circuit

metro/buses stand utilization and rebalancing



Module 3 - Modal Substitution & Transit Shift

How tourists reach and move through sites



Module 4 - Weather Sensitivity & Elasticity

Outdoor vs indoor demand shifts under different weather conditions

Technology Stack



Python/Google Colab

data generation script tied to existing weather/monument data

Supabase

Hosts new Fact tables:

`fact_tourist_bookings,`
`fact_heritage_transport,`
`fact_modal_shift_weather`

Streamlit

Icore interactive platform providing web-based dynamic dashboarding and custom parameter controls

dim_weather

Reused directly from Milestone 2's Open-Meteo integration

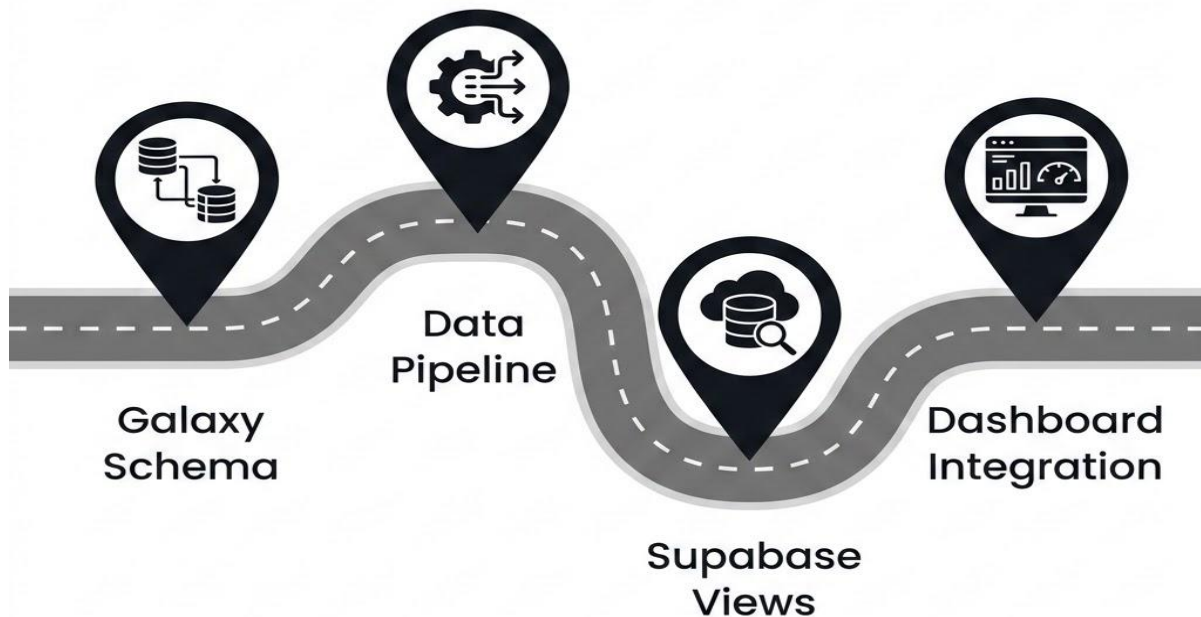
- ❏ Team uses a blended stack across specialized modules, targeted for unified integration in the final milestone.

About Our Datasets

Source	Coverage	Used For
fact_tourist_bookings	Simulated hourly bookings, fees, wait times, cancellations	Visit and Booking Intelligence
fact_heritage_transport	Simulated shuttle/e-rickshaw stand availability and trips	Last-Mile Circuit Analytics
fact_modal_shift_weather	Simulated mode-share distribution under different weather states	Modal Substitution Analysis
dim_weather	Historical monthly temperature, rainfall, humidity	Weather Sensitivity and Elasticity

❏ *Since detailed ticket and shuttle data doesn't exist publicly, we simulated it making sure it stays tied to the real weather and monument patterns we already have in our database.*

Methodology



Key Implementation Details

Galaxy Schema Extension

Integrated three core fact tables (fact_tourist_bookings, fact_heritage_transport, fact_modal_shift_weather) to expand the existing multidimensional schema

Integrated Data Pipeline

Standardized data ingestion workflows to process transaction-level tourist bookings, transit availability, and micro-climate indicators.

Supabase Views Layer

Utilized database views (view_booking_intelligence, view_transport_tourism_summary)

Streamlit Integration

Delivered a 4-module interactive web application featuring dynamic filters, geospatial maps, and predictive parameter toggles

Expected Outcomes



Hourly Demand Profiles

Revealing morning, midday, and evening peak windows



Transport Heatmaps

Last-mile heatmaps identifying stands needing vehicle rebalancing



Weather Elasticity

Measurable percentage demand shifts between outdoor and indoor sites



Booking Dashboards

Completion and cancellation dashboards flagging bottlenecked sites



Transit Mode Distribution

Showing how tourists actually reach and move through heritage sites



Dashboard Inventory

13 KPI cards and 12 visualizations, ready for Milestone 4's executive integration

Conclusion



Milestone 3 helps our platform move from just looking at tourism numbers to actually understanding how visitors behave on the ground. By putting bookings, queues, travel routes, and weather changes together in our data pipeline, we've built a solid operational setup. This gets everything ready for predictive modeling and executive dashboards in Milestone 4

Milestone 1

Galaxy Schema & Clean Data
Foundation

Milestone 2

Geospatial, Weather & Interactive
Dashboards

Milestone 3

Demand Intelligence and Transit
Mobility

Milestone 4

Predictive Forecasting & Executive
Synthesis